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Park et al.

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(54) **SEMI-CONDUCTOR DEVICE HAVING
DOUBLE-GATE AND METHOD FOR
SETTING SYNAPSE WEIGHT OF TARGET
SEMI-CONDUCTOR DEVICE WITHIN
NEURAL NETWORK**

USPC 365/149
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2005/0285184 A1* 12/2005 Jung H10D 30/69
257/324
2006/0267134 A1* 11/2006 Tilke H10B 41/30
257/E21.546
2009/0127613 A1* 5/2009 Ikehashi H10B 69/00
438/257

(Continued)

FOREIGN PATENT DOCUMENTS

KR 10-2015-0014577 A 2/2015
KR 10-2018-0127276 A 11/2018

OTHER PUBLICATIONS

H. Kim et al., "Silicon-Based Floating-Body Synaptic Transistor with Frequency-Dependent Short-and Long-Term Memories," IEEE Electron Device Letters, vol. 37, No. 3, pp. 249-252 (Mar. 2016).

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(57) **ABSTRACT**

Embodiments relate to a semiconductor device including a body made of a first conducting semiconductor material, a source and a drain made of a second conducting semiconductor material and formed on the body, a first gate formed on the body with a gate insulating layer interposed between the first gate and the body, a second gate formed opposite the first gate with respect to the body, and an insulating layer stack having a charge storage layer formed between the body and the second gate, and a method for controlling a synapse weight of a target semiconductor device within a neural network including semiconductor devices.

14 Claims, 10 Drawing Sheets

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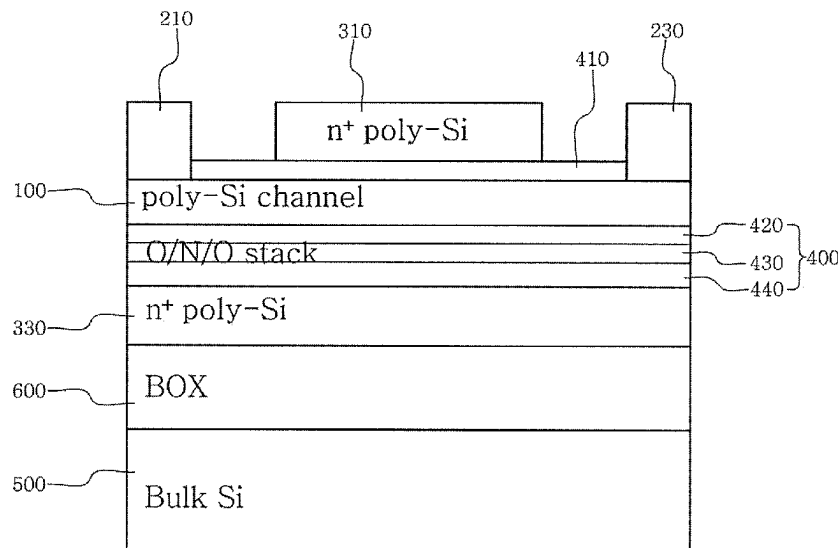
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G11C 16/12 (2006.01)
G11C 16/14 (2006.01)
H10B 43/30 (2023.01)
H10D 89/60 (2025.01)

(52) **U.S. Cl.**

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CPC G11C 16/0466; G11C 16/12; G11C 16/14;
H10B 43/30; H10D 89/601; G06N 3/063



(56)

References Cited

U.S. PATENT DOCUMENTS

2010/0102372 A1* 4/2010 Lee G11C 11/404
257/E27.084

* cited by examiner

FIG. 1

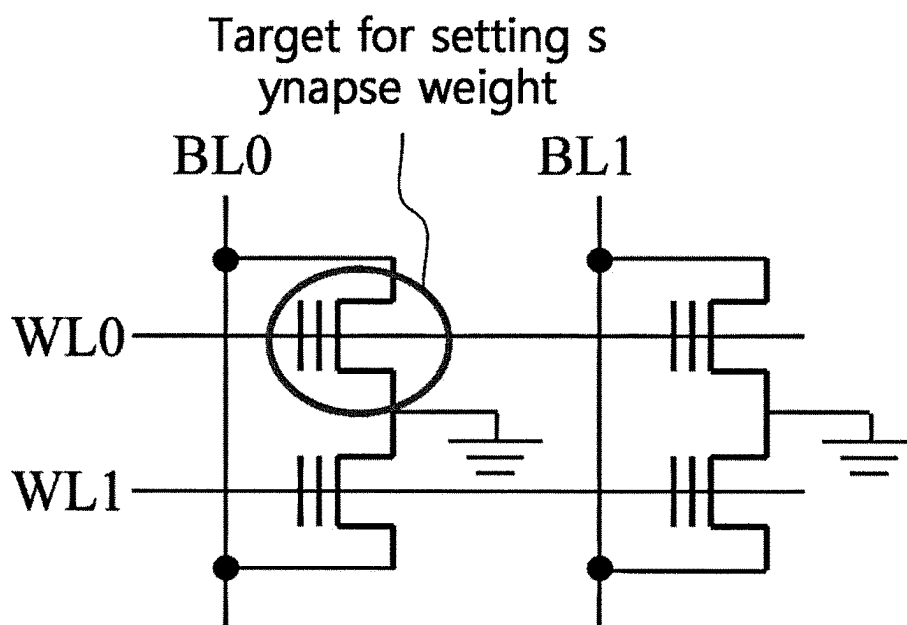


FIG. 2

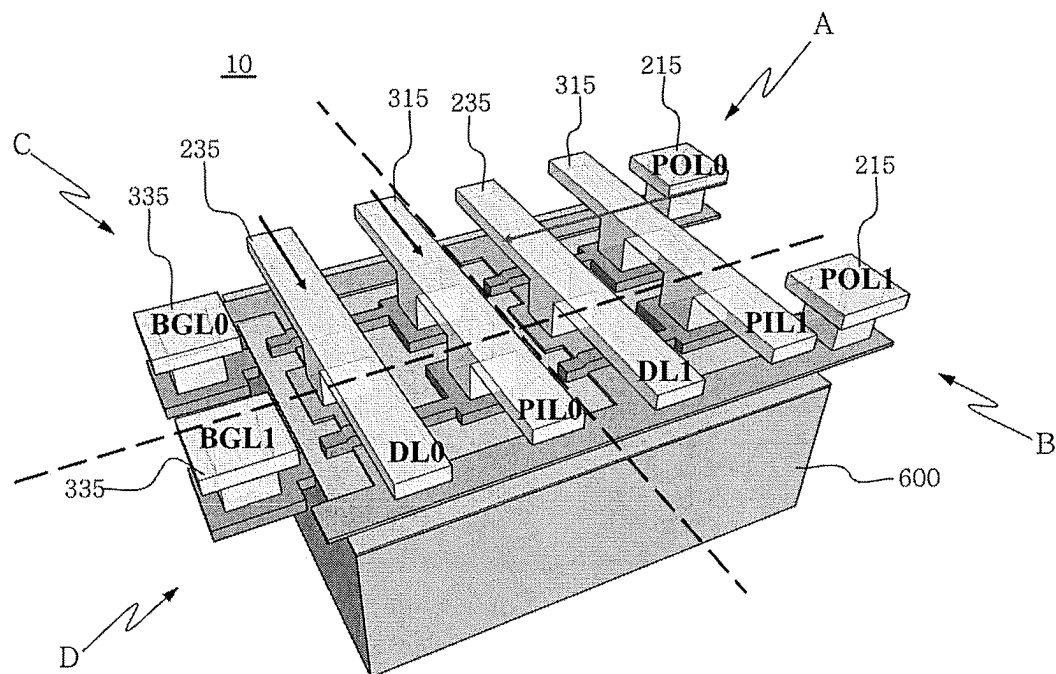


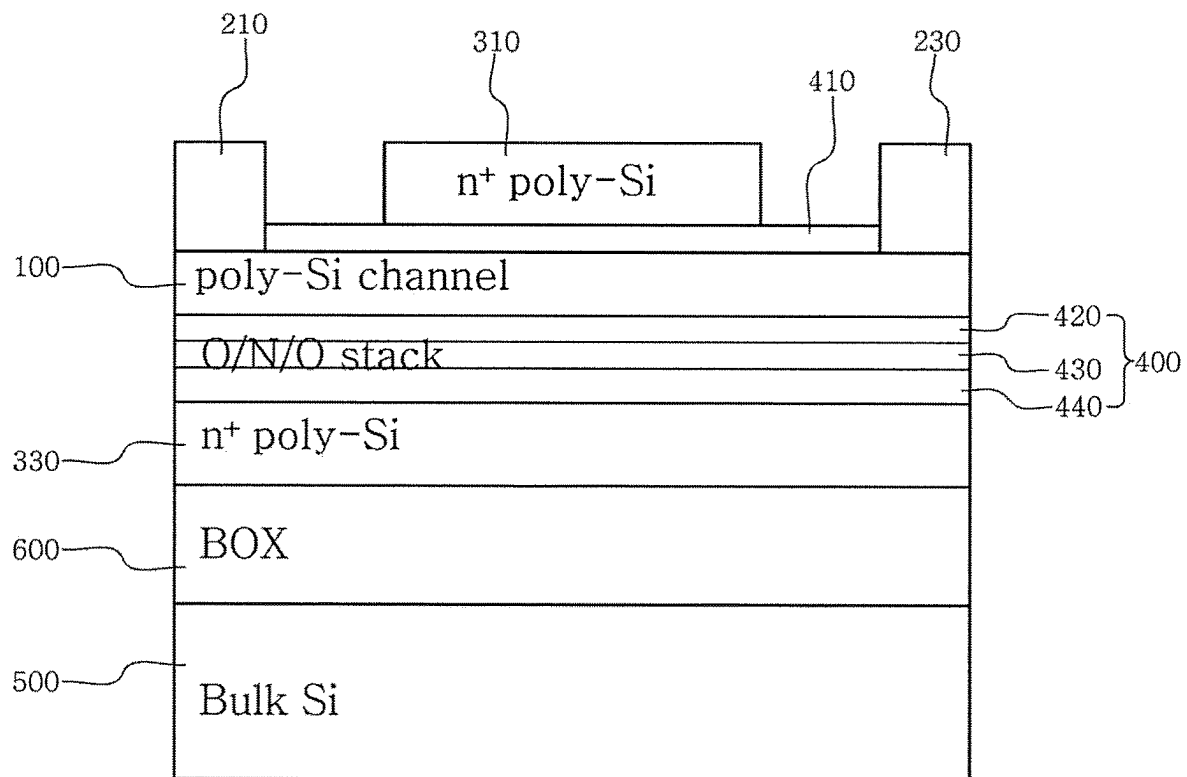
FIG. 3

FIG. 4

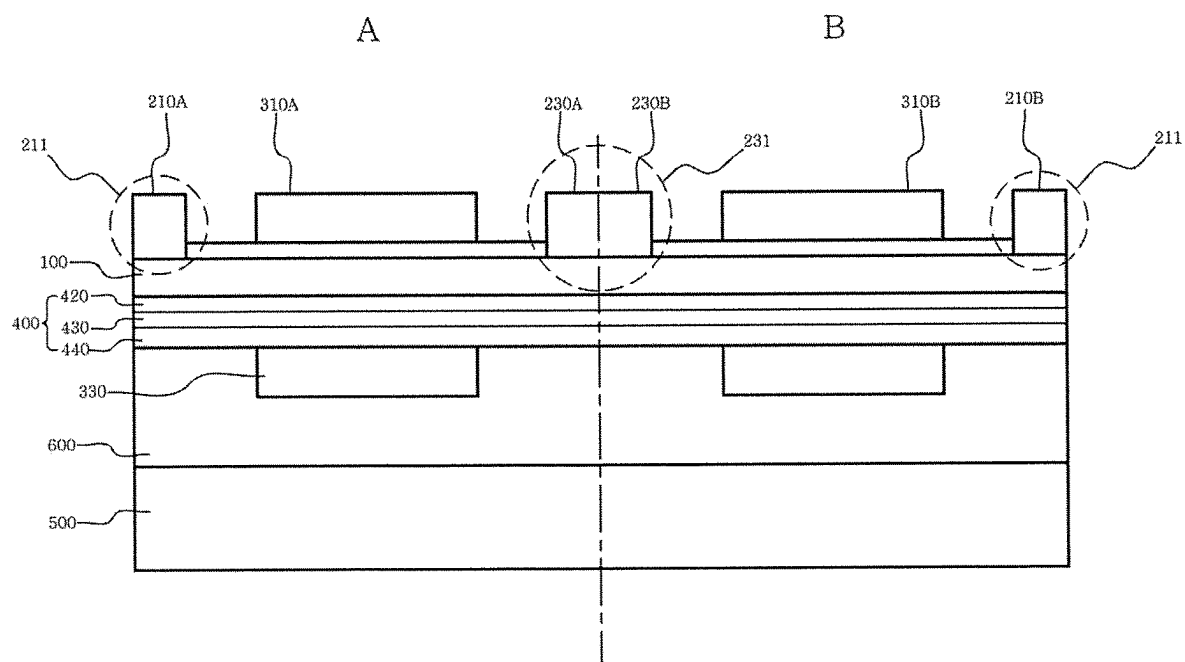


FIG. 5

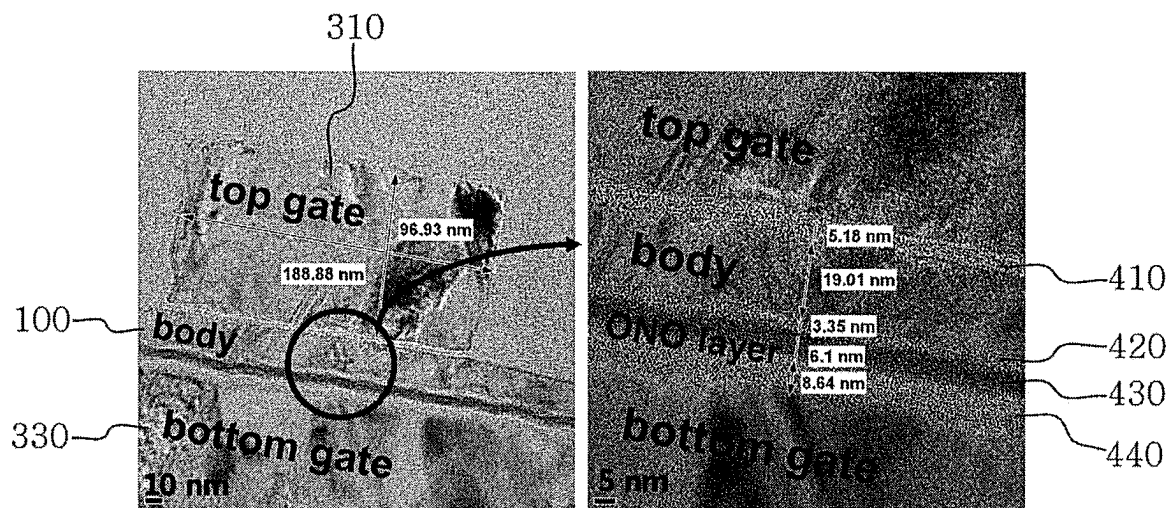


FIG. 6

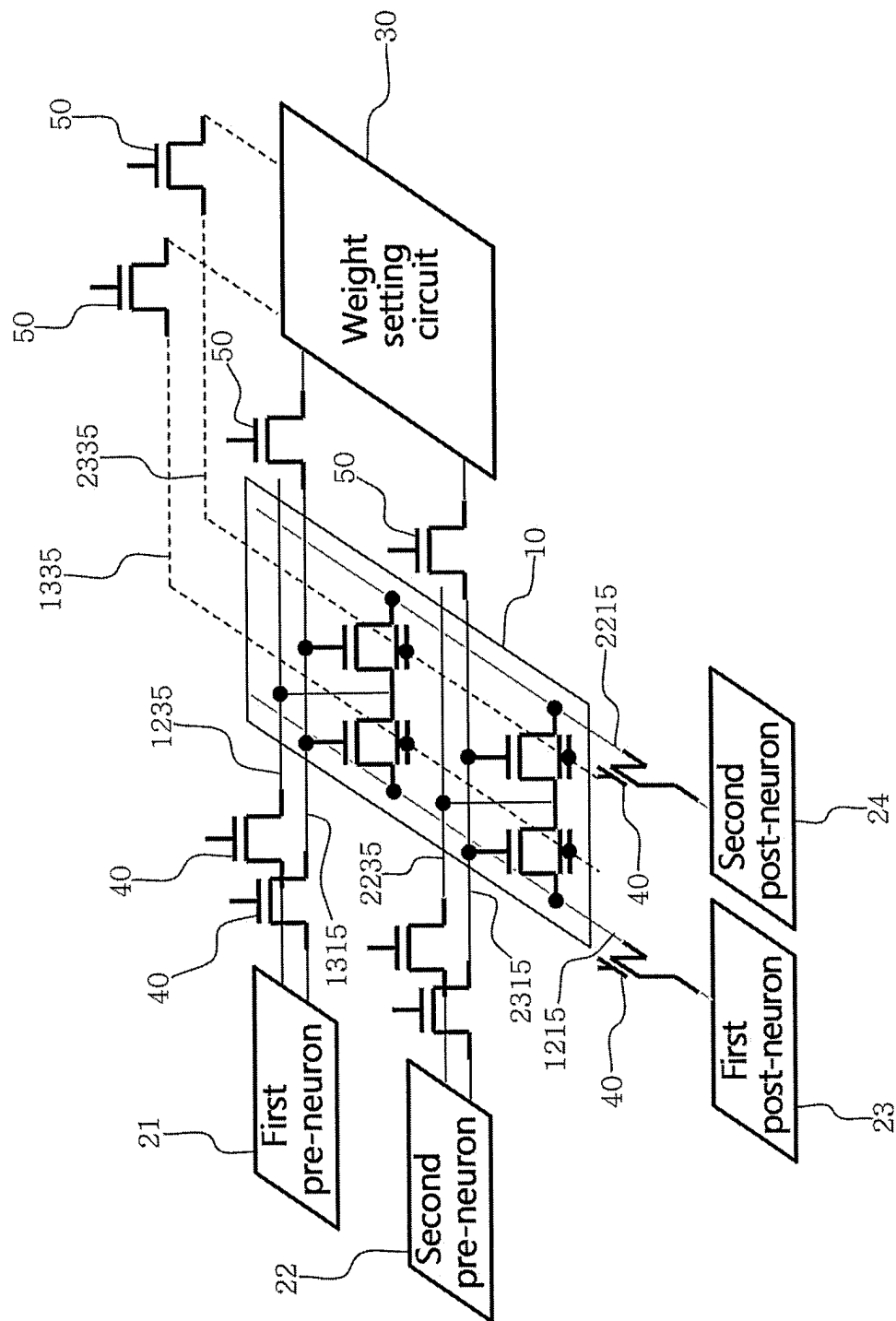


FIG. 7

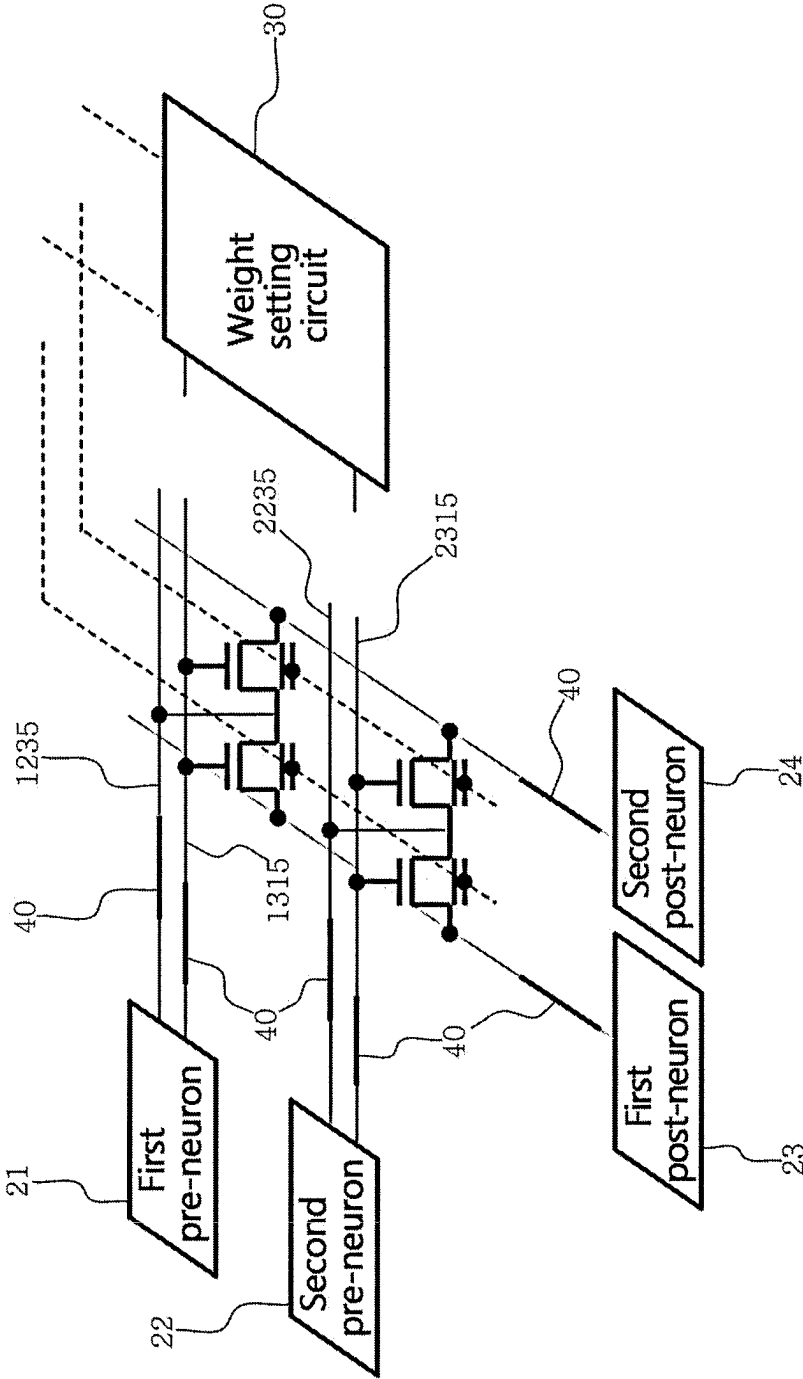


FIG. 8

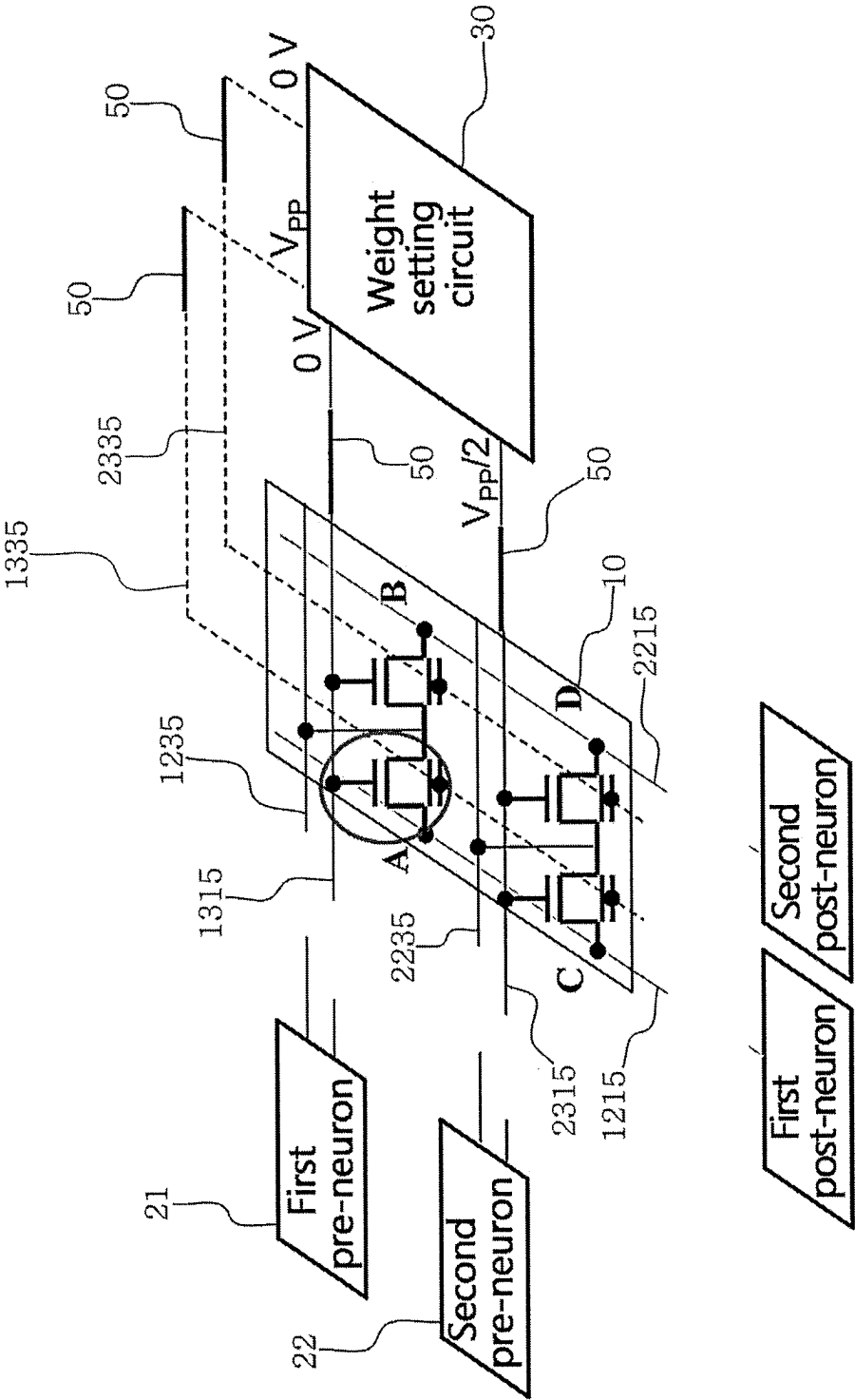


FIG. 9

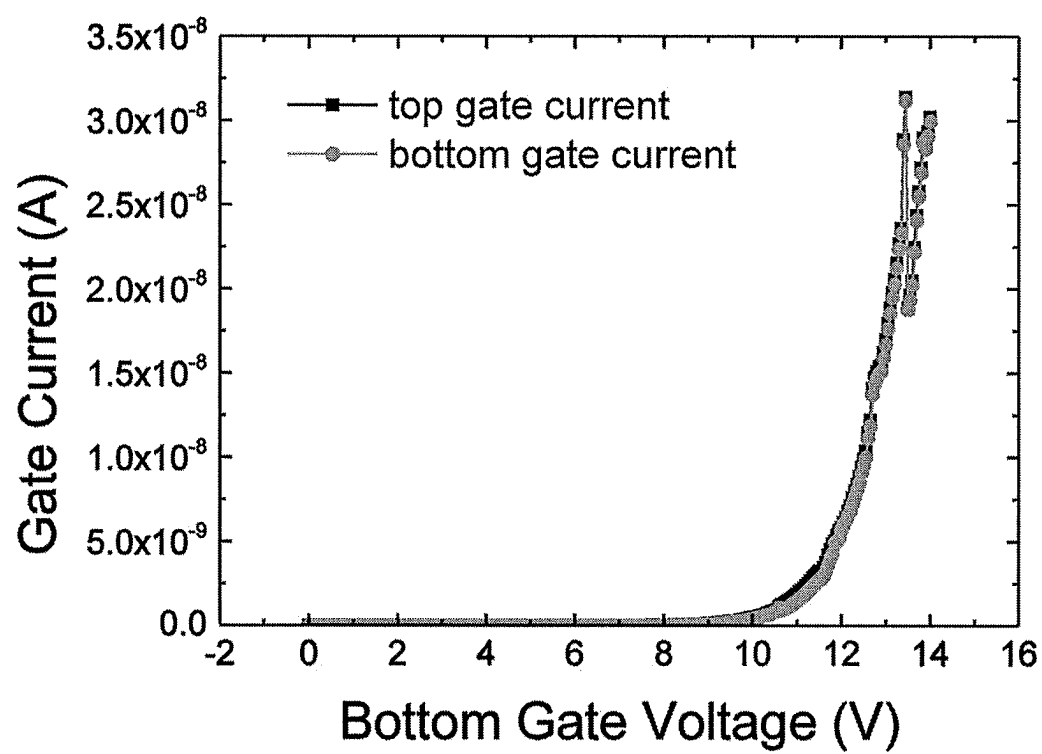
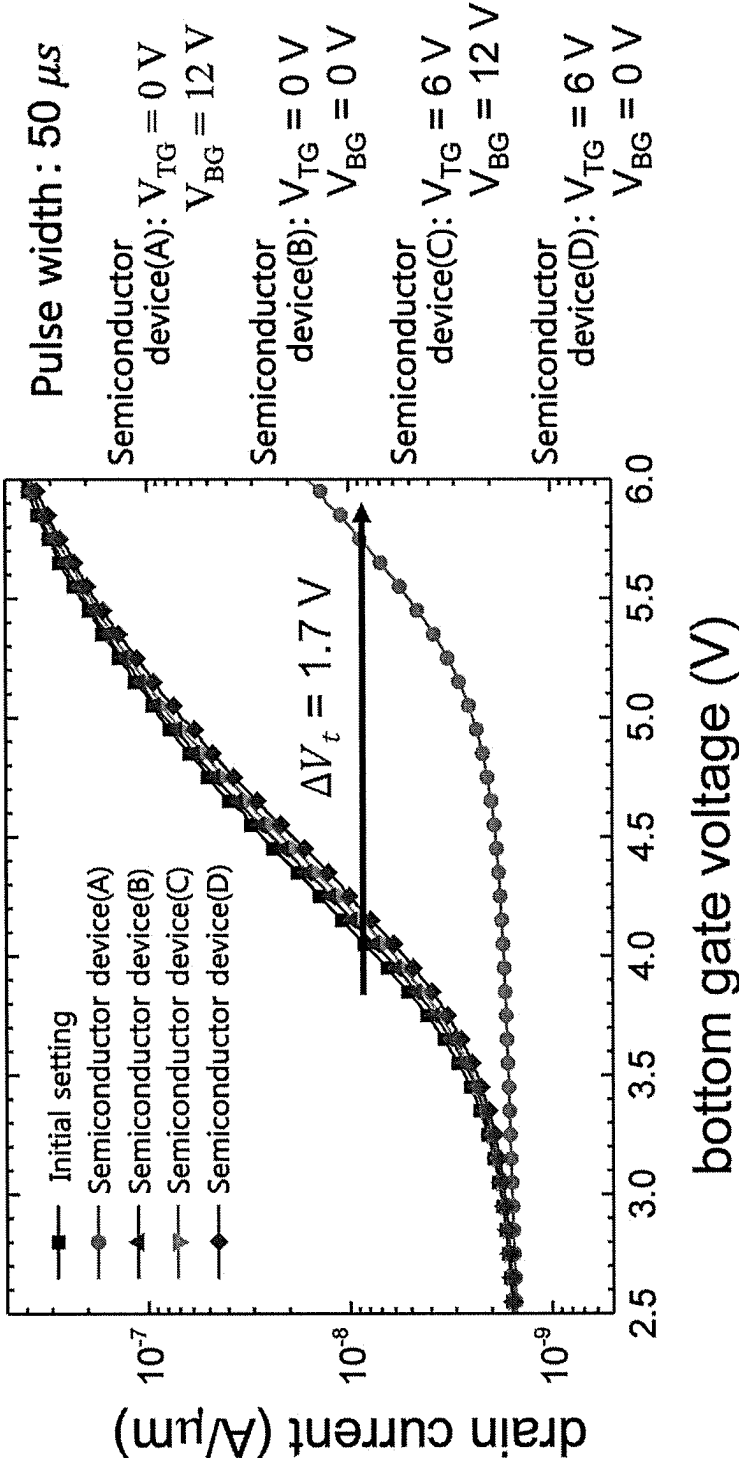


FIG. 10



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SEMI-CONDUCTOR DEVICE HAVING DOUBLE-GATE AND METHOD FOR SETTING SYNAPSE WEIGHT OF TARGET SEMI-CONDUCTOR DEVICE WITHIN NEURAL NETWORK

FIELD OF THE INVENTION

Embodiments relate to a semiconductor device for a hardware based neural network, and more particularly, to a semiconductor device which may be positioned in an array structure of a NOR flash memory and is configured to perform a program operation through Fowler-Nordheim (FN) tunneling and a method for independently setting a synapse weight of a semiconductor device within a neural network that includes these semiconductor devices.

NATIONAL RESEARCH AND DEVELOPMENT SUPPORT

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BACKGROUND OF THE INVENTION

Recently, there have been rapid developments in artificial intelligent related technology that emulates the functions of synapses and neurons of the biological nervous system. Many technologies that implement neural networks including synapses and neurons in software have been developed and are being developed. Additionally, technologies that implement the neural networks in hardware are also being vigorously developed.

The neural network implemented in hardware (hereinafter, "hardware neural network") includes a synapse array having a plurality of cells. To accurately design the hardware neural network, it is necessary to independently set the weight for each cell. To this end, it is necessary to independently control the weight of the hardware neural network for each cell.

In general, a group of device candidates for use as synapses include static random-access memory (SRAM), resistance change memory (for example, resistive random access memory (ReRAM)), phase change memory (PCM), spin torque transfer magnetic random access memory (STT_MRAM) and flash memory. Among them, the flash memory has non-volatile properties and very good characteristics in terms of reliability.

The flash memory may be classified into NOR flash memory and NAND flash memory according to the memory array structure. The NOR flash memory is a flash memory having an array structure in which a source region and a drain region are arranged in an alternating manner. The NOR flash memory is able to perform the weighted sum of synapses by reading source currents of unit synapses (for example, synapse blocks) since each cell is connected in parallel in the memory array as opposed to the NAND flash memory. Accordingly, to build the hardware neural network, it is effective to use the NOR flash memory.

Currently, the g NOR flash memory uses a channel hot electron (CHI) process for a program (PGM) operation for setting the weight. However, the CHI process requires a high drain current for the program (PGM) operation, which makes it difficult to manufacture the devices using polysili-

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con, so single-crystalline silicon is used to manufacture the devices, requiring high power consumption.

Meanwhile, a Fowler-Nordheim (FN) tunneling process is another way of the program operation. However, prior embodiments applying the FN tunneling process to the NOR flash memory has limitations for independently performing the program (PGM) operation for each cell.

FIG. 1 is a diagram for describing a prior problem faced when attempting to perform the program (PGM) operation on a target cell of the NOR flash memory through the FN tunneling process.

The NOR flash memory has an array structure including a word line WL, a bit line BL and a common source line SCL. When the FN tunneling process is applied to set the weight of the single target cell in a 2x2 block, a program voltage is applied through the word line WL0 connected to the target cell. Thus, program inhibit does not take place in two cells sharing the word line WL0 by the common source line irrespective of the voltage of the bit line BL0. That is, programming operations of other cells excepting the target cell cannot be inhibited.

SUMMARY OF THE INVENTION

According to an aspect of the present disclosure, there is provided a semiconductor device that maintains an array structure of a prior NOR flash memory in which drains of semiconductor devices are arranged adjacent to each other, and performs program (PGM)/erase (ERS) operation through Fowler-Nordheim (FN) tunneling between double gates.

In addition, the present disclosure further provides a method for independently setting a synapse weight of a target semiconductor device in a synapse block that includes semiconductor devices through the FN tunneling process.

A semiconductor device according to an aspect of the present disclosure may include a body made of a first conducting semiconductor material; a source and a drain that are made of a second conducting semiconductor material formed on the body; a gate insulating layer formed on the body; a first gate formed on the gate insulating layer; a second gate formed opposite the first gate, wherein the body is positioned between the first gate and the second gate; and an insulating layer stack having a charge storage layer formed between the body and the second gate.

In an embodiment, electrons of the first gate may tunnel into the second gate on a state where the source and the drain are floated if an electric field from the first gate to the second gate is formed by a potential difference between the first gate and the second gate.

In an embodiment, the insulating layer stack may further include a first stack insulating layer formed on the charge storage layer, and a second stack insulating layer formed opposite the first stack insulating layer. Here the charge storage layer is positioned between the first stack insulating layer and the second stack insulating layer.

In an embodiment, the first stack insulating layer may be configured to allow electric charges of the first gate to tunnel to the second gate if the electric field is formed by a potential difference between the first gate and the second gate. Here the potential difference between the first gate and the second gate causes to maintain each insulating layer without damaging each insulating layer.

In an embodiment, the first stack insulating layer may have a thickness of less than 7 nm.

In an embodiment, the first stack insulating layer may be further configured to prevent the electric charge tunneled in the charge storage layer from leaking when the electric field disappears.

In an embodiment, the first stack insulating layer may have a thickness of 3 nm or more.

In an embodiment, the second stack insulating layer may be configured to inhibit tunneling of electric charges of the first gate to the second gate if an electric field is formed by a potential difference between the first gate and the second gate. Here the potential difference between the first gate and the second gate causes to maintain each insulating layer without damaging each insulating layer.

In an embodiment, the second stack insulating layer may have a thickness of 8 nm or more.

In the above-described embodiments, the body may be made of any one of polysilicon, amorphous silicon and a combination thereof.

A neural network according to another aspect of the present disclosure has a plurality of semiconductor devices of the above mentioned embodiments as synapse cells. The neural network may be comprise a synapse block including a first synapse array and a second synapse array; drain lines; first gate lines; source lines. Here, the first synapse array includes a first semiconductor device and a second semiconductor device and a second synapse array includes a third semiconductor device and a fourth semiconductor device, each synapse array configured such that drains of different semiconductor devices are arranged adjacent to each other. The drain lines are electrically connected to a drain region of each synapse array. The first gate lines are electrically connected to a first gate of each synapse array. The source lines are electrically connected to a source region of the first synapse array and a source region of the second synapse array. The drain line and the first gate line receive an input signal from a pre-neuron, and the source line outputs a calculation result at the synapse block as an output signal to a post-neuron.

In an embodiment, the synapse block is configured to form a source signal based on the input signal and a preset weight for each semiconductor device when receiving the input signal.

In an embodiment, the output signal may be a sum of the source signals for each semiconductor device.

In an embodiment, the synapse block may connect the pre-neuron and the post-neuron through first switching devices which switch on/off electrical connections of the pre-neuron, the post-neuron and the synapse block. The synapse block may be further connected to a control circuit through second gate lines electrically connected to a second gate of the first synapse array and a second gate of the second synapse array, and a second switching device which switches on/off an electrical connection between the control circuit and the synapse block, wherein the control circuit controls the semiconductor device through at least some of the plurality of lines.

In an embodiment, the output of the calculation result may be performed in an ON state of the first switching device and an OFF state of the second switching device, and the operation of the control circuit may be performed in an OFF state of the first switching device and an ON state of the second switching device.

A method for setting a synapse weight of a target semiconductor device in a synapse array including a plurality of semiconductor devices according to still another aspect of the present disclosure may include: controlling a potential difference between a first gate and a second gate of the target

semiconductor device to form an electric field for tunneling of electric charge in the target semiconductor device that has a preset weight through tunneling; controlling a voltage of a second gate of another semiconductor device to prevent tunneling of electric charge in the other semiconductor device, the another semiconductor device sharing a first gate line between the first gate of the target semiconductor device and a control circuit; and controlling a voltage of a first gate of other semiconductor device to prevent tunneling of electric charge in the another semiconductor device, the other semiconductor device sharing a second gate line between the second gate of the target semiconductor device and the control circuit.

In an embodiment, controlling the potential difference between the first gate and the second gate of the target semiconductor device may include applying a first voltage to the second gate of the target semiconductor device, and applying a second voltage to the first gate of the target semiconductor device. Here, the potential difference between the first gate and the second gate of the target semiconductor device forms a channel for tunneling of an electron of the first gate to the second gate.

In an embodiment, according to the method, controlling the voltage of the second gate of another semiconductor device may include applying a same voltage as the voltage of the first gate of the target semiconductor device to the second gate of the other semiconductor device.

In an embodiment, controlling the voltage of the first gate of the other semiconductor device may include applying a third voltage to the first gate of the other semiconductor device to reduce to below the potential difference between the first gate and the second gate of the target semiconductor device.

In an embodiment, the third voltage may be a voltage which brings a potential difference between the first gate and the second gate of the other semiconductor device to 40% to 60% of the potential difference between the first gate and the second gate of the first semiconductor device.

Advantageous Effects

The semiconductor device according to an aspect of the present disclosure includes double gates, and has an asymmetrical structure with respect to the body. With the above-described semiconductor device, it is possible to form an array structure of a NOR flash memory in which drains of semiconductor devices within the same array are arranged adjacent to each other. Here, lines which control the drains are configured in contact with each other to share the adjacent drains within the same array.

The semiconductor device may perform the program operation through Fowler-Nordheim (FN) tunneling between the double gates. In particular, under the above-described array structure, it is possible to independently set the weight of the target semiconductor device through FN tunneling.

When the program (PGM) operation is performed through the FN tunneling process, it is possible to reduce power consumption to about 1/1000 of that of the channel hot electron (CHI) process.

As a result, using the semiconductor device, it is possible to form a low power/high integration synapse array, and further, to improve the performance compared to the existing NOR flash memory based neural networks.

Further, when polycrystalline semiconductor materials (such as, for example, polysilicon) commonly used in the field of semiconductors are used for the body of the semi-

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conductor device, it is easy to expand to an array structure, and there are advantages in terms of process complexity and integration.

BRIEF DESCRIPTION OF THE DRAWINGS

The following is a brief introduction to necessary drawings in the description of the embodiments to describe the technical solutions of the embodiments of the present disclosure or the existing technology more clearly. To identify similar elements shown in at least one drawing, a same reference numeral is used. It should be understood that the accompanying drawings are for the purpose of describing the embodiments of the present disclosure and are not intended to be limiting of the present disclosure. Additionally, for clarity of description, illustration of some elements in the drawings may be exaggerated and omitted.

FIG. 1 is a diagram for describing a problem faced when attempting to perform a program (PGM) operation on a target cell of an existing NOR flash memory through a Fowler-Nordheim (FN) tunneling process.

FIG. 2 is a conceptual perspective view of a synapse block having a synapse array including a plurality of semiconductor devices according to an embodiment of the present disclosure.

FIG. 3 is a conceptual cross-sectional view showing a semiconductor device according to an embodiment of the present disclosure.

FIG. 4 is a cross-sectional view of a synapse array included in a synapse block according to an embodiment of the present disclosure.

FIG. 5 is a transmission electron microscopy (TEM) image showing a cross section of a semiconductor device according to an embodiment of the present disclosure.

FIG. 6 is a circuit diagram showing a neural network including a synapse block according to an embodiment of the present disclosure.

FIG. 7 is a diagram showing a connection configuration of the neural network of FIG. 6 in an inference mode.

FIG. 8 is a diagram for describing a process of adjusting a synapse weight of a target cell in a synapse block of FIG. 5.

FIG. 9 is a diagram showing the characteristics of the target semiconductor device of FIG. 8 in a synapse weight setting process.

FIG. 10 is a diagram showing the result according to the synapse weight setting method of FIG. 8.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Hereinafter, the embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

When an element is referred to as being “on” another element, the element may be on the other element, or intervening elements may be interposed between. In contrast, when an element is referred to as being “immediately on” another element, there is no intervening element between them.

The terms “first”, “second”, “third” and the like are used to describe various portions, components, regions, layers and/or sections, but are not limited thereto. These terms are used to distinguish a portion, component, region, layer or section from another portion, component, region, layer or section. According, a first portion, component, region, layer or section described herein may be referred to as a second

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portion, component, region, layer or section without departing from the range of the present disclosure.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “comprises” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements and/or components.

The terms indicating relative spaces such as “below”, “on” and the like may be used to describe a relationship of an element to another element shown in the drawing more easily. These terms are intended to include not only the intended meanings in the drawings but also other meanings or operations of a device used. For example, when the device in the drawing is reversed, elements described as being “below” other elements are understood as being “on” the other elements. Accordingly, the exemplary term “below” includes both up and down directions. The device may rotate at 90° or different angles, and the terms indicating relative spaces are interpreted accordingly.

Unless otherwise defined, all terms including technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

A semiconductor device according to embodiments of the present disclosure may form an array structure having a similar structure to a NOR flash memory. Each semiconductor device in the array structure may independently perform a write (or program (PGM) and/or erase (ERS)) operation through Fowler-Nordheim (FN) tunneling.

In an embodiment, the semiconductor device acts as a synapse cell for forming a hardware based neural network. The neural network includes at least one synapse block, and the synapse block includes a plurality of synapse cells. In the specification, the synapse cell refers to the semiconductor device used in the neural network, and thus the terms “semiconductor device” and “synapse cell” are interchangeably used herein. Additionally, since a series of semiconductor devices are arranged, the terms “memory array” and “synapse array” are also interchangeably used herein. Semiconductor Device and Synapse Array

FIG. 2 is a conceptual perspective view of a synapse block having a synapse array including a plurality of semiconductor devices according to an embodiment of the present disclosure.

Referring to FIG. 2, the synapse block 10 includes a plurality of semiconductor devices. The synapse block 10 has a synapse array in which at least one semiconductor device is arranged.

In an embodiment, the synapse block 10 includes a first synapse array including a first semiconductor device A and a second semiconductor device B. Additionally, in some embodiments, the synapse block 10 includes a second synapse array including a third semiconductor device C and a fourth semiconductor device D. The first synapse array and the second synapse array are arranged in parallel.

Hereinafter, the embodiments of the present disclosure will be described in detail using the synapse block **10** of 2×2 semiconductor devices. However, it will be obvious to those skilled in the art that the configuration of the synapse block **10** of the present disclosure is not limited thereto.

FIG. **3** is a conceptual cross-sectional view showing the semiconductor device according to an embodiment of the present disclosure.

Referring to FIG. **3**, the semiconductor device of the synapse block **10** includes: a body **100**; a source **210** and a drain **230**; a first gate **310**; a gate insulating layer **410**; a second gate **330**; and an insulating layer stack **400** having a charge storage layer **430**.

The body **100** is made of a first conducting (for example, p-type) semiconductor material. Both of the source **210** and the drain **230** are made of a second conducting (for example, n-type) semiconductor material.

The source **210** and the drain **230**, and the gate insulating layer **410** are formed on a surface of the body **100**. The source **210** and the drain **230** are positioned in contact with the body **100** and are spaced apart from each other. The first gate **310** is formed the gate insulating layer **410**.

The body **100** is positioned between the first gate **310** and the second gate **330**. The insulating layer stack **400** is formed between the body **100** and the second gate **330**.

In some embodiments, the semiconductor device may be formed on a construction that includes a substrate **500** and an embedded oxide layer **600**.

In the specification, the first gate **310** may be referred to as a top-gate, and the second gate **330** may be referred to as a bottom-gate because the first gate **310** is disposed at a higher position than the second gate **330** with respect to the substrate **500**, shown in the FIG. **3**.

The body **100** may be formed from a material selected from the group consisting of polycrystalline semiconductor materials in which grain boundaries are clearly defined, including polysilicon or polycrystalline germanium, but is not limited thereto. Alternatively, the body **100** may be formed from a material selected from the group consisting of amorphous semiconductor materials. The body **100** is formed from a polycrystalline or amorphous semiconductor material instead of a single-crystalline semiconductor material, thereby allowing 3D stacking.

The semiconductor device has the double gate **310**, **330**. The body **100** is located between the double gate **310**, **330**. The semiconductor device is configured to store electric charge in the charge storage layer **430** of the insulating layer stack **400**. When an electric field is formed in the semiconductor device, electric charge is stored in the charge storage layer **430**. The electric charge stored in the charge storage layer **430** is electrically isolated to prevent leaks after the electric field disappears. In an embodiment, as shown in FIG. **3**, the insulating layer stack **400** may include insulating layers **420**, **440**. The charge storage layer **430** is disposed between the insulating layers **420**, **440**.

In an embodiment, the double gates **310**, **330** may be made of a polycrystalline semiconductor material including polysilicon. Additionally, the double gates may be made of a first conducting semiconductor material (for example, n-type materials).

FIG. **4** is a cross-sectional view of the synapse array included in the synapse block according to an embodiment of the present disclosure.

In the neural network, the program/erase operation is performed for each synapse block. The programming/erasing method of the synapse block **10** depends on the array structure of the semiconductor device.

Referring to FIG. **4**, the synapse array includes at least two semiconductor devices, and has an array structure in which drains of semiconductor devices are arranged adjacent to each other.

For example, in the synapse array having first and second semiconductor devices A, B, a drain region **231** includes drains (for example, **230A**, **230B**) of the different semiconductor devices A, B. Additionally, a source region **211** of the synapse array includes sources **210A**, **210B** of the semiconductor devices A, B.

The adjacent drains (for example, **230A**, **230B**) share a line through the drain region **231**. In FIG. **2**, the drain line **235** is connected to the drains **230A**, **230B** at the drain region **231**.

In some embodiments, the synapse array may be further configured in which sources of different semiconductor devices are also arranged adjacent to each other. For example, when the synapse array includes at least three semiconductor devices A, B, C, sources (for example, **210B**, **210C**) of different semiconductor devices B, C are arranged. When the synapse array in which sources are arranged adjacent to each other is used in the neural network, the adjacent sources (for example, **210B**, **210C**) do not share a line and are connected to separate lines and each configured to output a calculation signal for an input signal of the post-neuron. Referring FIGS. **2** and **4**, the sources **210B**, **210C** are connected to the different source lines **215** respectively.

The synapse array in the synapse block **10** in the embodiments (for example, shown in FIG. **4**) is similar to the array structure of the prior NOR flash memory. But the synapse array is configured to work in an operation principle that is different from the operation principle of the prior NOR flash memory, as described below.

The prior NOR flash memory performs the program operation through a channel hot electron (CHI) process. The CHI process applies a high voltage (for example, 12V to 14V) to a control gate, and applies a voltage to a drain, the voltage is high but lower than the voltage of the control gate. Accordingly, a semiconductor device designed to perform the program/erase operation through the CHI process requires a relatively high drain current. When the body **100** is formed from a polycrystalline or amorphous semiconductor material, it is difficult to perform the program operation through the CHI process due to the lower allowable current than a single-crystalline semiconductor material.

In contrast, the semiconductor device of the synapse block **10** has a similar array structure to the array structure of the existing NOR flash memory, but performs the program and/or erase operation through a FN tunneling process rather than the CHI process.

Program Operation

Referring back to FIG. **2**, the program operation of the semiconductor device will be described.

The semiconductor device of the present disclosure performs the program operation by forming a channel from the top gate **310** to the bottom gate **330** through FN tunneling. The synapse weight of the semiconductor device is adjusted through the program operation.

In an embodiment, an electric field from the top gate **310** to the bottom gate **330** is formed. The electric field is formed by applying a higher voltage to the bottom gate **330** and a lower voltage to the top gate **310**. For example, the electric field is formed by controlling the voltage of the top gate **310** as 0V and the voltage of the bottom gate **330** as a positive value. A potential difference between the bottom gate **330** and the top gate **310** for forming the electric field depends

on the material properties of the body **100**, the thickness of the body **100**, the material properties of the insulating layer **410**, **420** or **440** and the thickness of the insulating layer **410**, **420** or **440**.

The potential difference for forming the electric field may be referred to as a program potential difference. Accordingly, when the voltage of the top gate **310** is 0V, the voltage of the bottom gate **330** is referred to as a program voltage.

As described above, when the program voltage is applied to the bottom gate **330**, a channel is formed between the double gates **310**, **330**, and electric charge is stored in the charge storage layer **430**. Here, the program voltage is any voltage of predetermined range. A minimum voltage value of the program voltage may be a lowest one of a voltage that makes to the stored electric charge from leaking out of the charge storage layer **430** in the semiconductor device having the insulating layer **410**, **420**, or **440** when the electric field disappears and another voltage for tunneling electric charge of the top gate **310** into the insulating layers **410**, **420** between the top gate **310** and for tunneling the charge storage layer **430**. A maximum voltage value of the program voltage may be a voltage for not damaging each insulating layer **410**, **420**, **440** when the corresponding voltage is applied.

Hence, the semiconductor device is designed to enable FN tunneling inside.

The insulating layers **410**, **420** are so thin that the electric charge of the top gate **310** can tunnel into the bottom gate **330** when voltage within the allowable voltage range of the body **100** (for example, made of a polycrystalline or amorphous semiconductor material) is applied. When the insulating layer **410** of the top gate **310** and/or the insulating layer **420** of the stack **400** are too thick, a tunneling current from the top gate **310** reduces, thereby failing to adjust the weight.

The existing NOR flash memory in which voltage of 10V or 12V is applied as the gate voltage uses a tunneling insulating layer having the thickness of 7 nm, 10 nm and so on. However, when the tunneling insulating layers **410**, **420** of FIG. 3 is designed to have the thickness of 7 nm, 10 nm, FN tunneling does not occur.

Accordingly, the insulating layer **410** is thinner than the gate insulating layer (for example, 7 nm to 10 nm) of the common transistor.

On the other hand, the insulating layers **420**, **440** are thick enough to store the electric charge of the charge storage layer **430** when there is no electric field. When the insulating layers **420**, **440** are too thin (for example, less than 3 nm), there is a high likelihood that a leak current may occur due to leakage of the electric charge in the charge storage layer **430** when performing the read operation through the top gate **310**.

Additionally, the insulating layer **440** is thicker than the insulating layers **410**, **420** to inhibit tunneling from the charge storage layer **430** to the bottom gate **330** under the electric field generated in the insulating layers **410**, **420**. For example, the insulating layer **440** is designed with a thickness for inhibiting FN tunneling into the bottom gate **330** under the allowable maximum voltage of the body **100**.

FIG. 5 is a transmission electron microscopy (TEM) image showing the cross section of the semiconductor device according to an embodiment of the present disclosure.

In an embodiment, the insulating layer **410** or **420** may have the thickness of less than 7 nm for movement of electric charge through FN tunneling. Additionally, the insulating layer **410** or **420** has the thickness of 3 nm or more

to prevent leakage of electric charge when the electric field disappears. Meanwhile, the insulating layer **440** is thicker than the insulating layers **410** and **420** to prevent the movement of electric charge at the voltage at which tunneling occurs in the insulating layers **410** and **420**.

In some embodiments, the gate insulating layer **410** is designed to have the thickness of 5 nm to 6 nm. Additionally, the insulating layer **420** of the stack **400** is designed to have the thickness of 3 nm to 4 nm. Additionally, the insulating layer **440** between the charge storage layer **430** and the bottom gate **330** is designed to have the thickness of 8 nm to 9 nm.

For example, as shown in FIG. 5, the insulating layers **410**, **420** are 5.18 nm and 3.35 nm in thickness, respectively, and the insulating layer **440** is 8.64 nm in thickness.

When the program voltage (for example, 10V to 15V) is applied to the bottom gate **330** of the semiconductor device having the insulating layers **410**, **420** that have the above-described thickness, a channel is formed between the gates **310**, **330** and electric charges of the top gate **310** pass the insulating layer **410** and electric chargers moves to the charge storage layer **430** through tunneling into the insulating layer **420**. The insulating layer **440** inhibits the tunneling electric chargers into the bottom gate **330**.

On the other hand, when the body **100** is too thick, the electric charge of the top gate **310** may not move to the charge storage layer **430** even though a predetermined range of voltage is applied. Accordingly, the body **100** is thick enough to allow the electric charge of the top gate **310** to move to the charge storage layer **430** through the insulating layers **410**, **420** under the allowable electric field. For example, when the body **100** is made of polysilicon, the body **100** may be 20 nm or less in thickness.

As described below with reference to FIG. 8, the semiconductor device according to the embodiments of the present disclosure may form a similar synapse array to the array structure of a NOR flash memory and store electric charge in the charge storage layer **430** through the FN tunneling process.

Erase Operation

In the similar way to the program operation, the erase operation of the semiconductor device is also performed by FN tunneling. However, an electric field from the bottom gate **330** to the top gate **310** is formed to cause FN tunneling for the erase operation. That is, a potential difference is set in the opposite direction to the program operation.

In the above example, the erase operation may be performed by releasing stored electrons from the charge storage layer **430** through FN tunneling or storing holes in the charge storage layer **430** through FN tunneling, by applying a lower voltage than the voltage of the top gate **310** (for example, a negative voltage having the same absolute value as the positive voltage for the program operation) to the bottom gate **330**.

The structure of the semiconductor device is not limited to FIG. 2. For example, the nitride layer stack **400** may be disposed in the top gate **310** direction. In this case, the semiconductor device performs the read operation through the bottom gate **330**.

The semiconductor device of FIG. 2 may be used as a single device that performs programming and/or erasing through FN tunneling. However, it is not limited thereto and may be expanded to a flash memory array.

In an embodiment, the flash memory array may be used to form the neural network as the synapse array.

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Synapse Block and Neural Network

The synapse block **10** including the above-described semiconductor device is configured to independently perform the program and/or erase operation for each cell, and thus it is possible to independently set a synapse weight for each cell. Additionally, the weighted sum calculation may be performed simply by reading the source current of the synapse block.

FIG. **6** is a circuit diagram showing the neural network including the synapse block according to an embodiment of the present disclosure.

Referring to FIG. **6**, the neural network **1** includes the synapse block **10** between neurons. According to the signal direction of the inference operation, a neuron transmitting a signal to the synapse block **10** is referred to as a pre-neuron of synapse, and a neuron receiving a signal from the synapse block **10** is referred to as a post-neuron of synapse.

The neurons and the synapse block **10** are connected with each other by the synapse array. As shown in FIG. **6**, when two synapse arrays of the synapse block **10** is included, pre-neurons **21**, **22** and post-neurons **23**, **24** are each connected to the synapse block **10**.

The neural network **1** includes a first switching device **40** to switch on/off the electrical connection between the synapse block **10** and the neurons **21**, **22**, **23**, **24**.

The synapse weight of the synapse block **10** is set by a separate control circuit (for example, a peripheral circuit, etc.) **30**. The control circuit **30** may be connected to the target synapse block **10** intended to set the weight in the neural network **1** to adjust and set the weight of a particular synapse cell of the synapse block **10**. The control circuit **30** may control the synapse block **10** through a plurality of control lines.

Additionally, the control circuit **30** and the synapse block **10** are further connected through a second switching device **50** which switches on/off the electrical connection. The adjustment of the synapse weight will be described in more detail below with reference to FIG. **8**.

The switching operation of the first and second switching devices **40**, **50** may be controlled by the control circuit **30** or a separate controller (for example, a computer, etc.).

Additionally, the first and second switching devices **40**, **50** may be a transistor as shown in FIG. **6**, etc., but are not limited thereto.

FIG. **7** is a diagram showing the connection configuration of the neural network of FIG. **6** in the inference mode.

When the neural network **1** performs the inference operation, the pre-neurons **21**, **22**, the synapse block **10** and the post-neurons **23**, **24** are electrically connected by the first switching device **40**. As shown in FIG. **7**, the synapse array is connected to each of the pre/post-neurons **21**, **22**, **23**, **24**.

On the contrary, the connection between the synapse block **10** and the control circuit **30** may be interrupted by the second switching device **50**.

The synapse block **10** receives input signals from the pre-neurons **21**, **22** and transmits calculation signals based on the input signals and the weight of the synapse block **10** to the post-neurons **23**, **24**.

Read Operation

Referring back to FIGS. **2** to **4**, the operation of the synapse block **10** in the inference mode will be described in more detail.

The synapse block **10** of FIGS. **3** and **4** performs the read operation through the top gate **310**. The synapse block **10** receives the input signals from the pre-neurons **21**, **22** through the top gate **310** and/or the drain **230** (or the drain region **231**).

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In an embodiment, the neural network **1** includes a top gate line **315** connected to the top gate **310**. As the top gate line **315** is a line which receives the input signals from the pre-neurons, the top gate line **315** may be referred to as a pre-synaptic input line (PIL). When the synapse block **10** includes the first synapse array and the second synapse array, the neural network **1** includes two top gate lines **1315**, **2315**.

Additionally, the neural network **1** includes a drain line (DL) **235** connected to the drain **230** (or the drain region **231**). The drain line **235** also receives the input signals from the pre-neurons **21**, **22**. When the synapse block **10** includes the first synapse array and the second synapse array, the neural network **1** includes two drain lines **1235**, **2235**.

In an embodiment, the drain line **235** is positioned in parallel to the top gate line **315**.

In some embodiments, a drain plug **233** is connected to share drains (for example, **230A**, **230B**) of two cells to increase the area efficiency of the array. For example, as shown in FIG. **3**, the drain plug **233** is connected to the drain region **231** including the drains **230A**, **230B** of the two cells A, B, and is configured to come into contact with both the drains **230A**, **230B** of the two cells A, B.

When the synapse block **10** receives the input signals from the pre-neurons **21**, **22** through the top gate line **315** and the drain line **235**, the synapse block **10** generates output signals of the cells representing the multiplication of the input signals by a preset synapse weight for each cell, and transmits each of the generated output signals of the cells to the post-neurons **23**, **24**.

The neural network **1** includes a source line **215** connected to the source region **211**. The source line **215** transmits the source current representing the multiplication of the input signal by the synapse weight to the post-neurons **23**, **24**.

In an embodiment, the source line **215** is positioned perpendicular to the top gate line **315**. The source line **215** transmits, to the post-neurons, the parallel sum of output currents of the synapse array disposed in a direction perpendicular to the direction of the top gate line **315**. The output current of the synapse array represents the sum of respective source currents (i.e., the calculation signal) of the synapse cells included in the corresponding array.

As a result, the neural network **1** may automatically perform weighted sum calculation of the synapse block **10** through the source line **215**.

As the source current is the output signal of the synapse block **10** transmitted to the post-neurons **23**, **24**, the source line **215** may be referred to as a post-synaptic output line (POL). When the synapse block **10** includes the first synapse array and the second synapse array, the neural network **1** includes two source lines **1215**, **2215**.

Weight Setting for Each Synapse Cell

As described above, the semiconductor device within the synapse block **10** performs the program/erase operation through the potential difference between the double gates (for example, through the voltage control of the bottom gate **330** when the top gate **310** is 0V).

For the program/erase operation, in addition to some control lines **1235**, **2235**, **1315**, **2315** of the control lines for the read operation as described above, an additional control line is required.

Referring to FIGS. **2** and **6**, the neural network **1** includes a bottom gate line (BGL) **335** connected to the bottom gate **330** of the first and second synapse arrays. When the neural network **1** includes the first synapse array and the second synapse array shown in the FIG. **2**, the synapse block **10** includes two bottom gate lines **1335**, **2335**.

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Each cell within the synapse array is connected in parallel to the source line **215**, and the drain line **235** and the source line **215** are perpendicular to each other, similar to the array structure of a NOR flash memory.

By a method for setting the weight of the synapse block according to another aspect of the present disclosure, it is possible to adjust the synapse weight of a particular target cell of the synapse block **10** having a similar array structure to a NOR flash memory through the FN tunneling process. Further, it is possible to independently set the synapse weight for each cell of the synapse block **10** through the FN tunneling process.

FIG. **8** is a diagram for describing a process of adjusting the synapse weight of the target cell in the synapse block of FIG. **5**.

Referring to FIG. **8**, in the weight adjustment mode, as opposed to the inference mode, the synapse block **10** is disconnected from the neurons and connected to the control circuit **30** for weight adjustment through FN tunneling.

For clarity of description, the weight adjusting process is described under the assumption of an example of adjusting the synapse weight of the semiconductor device A in the synapse block **10** including the semiconductor devices A, B, C, D.

The semiconductor device A is a target cell for weight adjustment, and to perform the program operation of the semiconductor device A through FN tunneling, the source **210A** and the drain **230A** of the target semiconductor device A are allowed to float. In an embodiment, the source **210A** and the drain **230A** are allowed to float by the switching-off of the first switching device **40**.

To form a channel from the top gate **310A** of the semiconductor device A to the bottom gate **330A** of the semiconductor device A, a potential difference between the top gate **310A** and the bottom gate **330A** is controlled. For example, a channel is formed by applying different voltages to the top gate **310A** and the bottom gate **330A** of the target semiconductor device A to control a potential difference between the top gate **310A** and the bottom gate **330A**.

When an electric field from the top gate **310A** to the bottom gate **330A** is formed, electric charge of the top gate **310A** moves to the bottom gate **330A**. In an example, the control circuit **30** controls the voltage of the top gate **310A** to 0V through the top gate line **1315** connected to the semiconductor device A, and applies the program voltage (V_{PP}) through the bottom gate line **1335** connected to the semiconductor device A to move electric charge of the bottom gate **310B** to the charge storage layer **430A** through FN tunneling. The program voltage (V_{PP}) may be, for example, 12V as described above, but is not limited thereto.

Accordingly, a channel of electric current from the top gate **310A** to the bottom gate **330A** is formed, and electric charge tunneled in the top gate **310A** moves to the bottom gate **330A** through tunneling. Finally, the tunneled electric charge is trapped in the charge storage layer **430** by the nitride layer **440**, thereby adjusting the weight of the target cell A.

Additionally, the remaining synapse cells B, C, D other than the target cell are controlled to program inhibit.

The semiconductor device B is a synapse cell that shares the top gate line **1315** with the target semiconductor device A, and the same voltage as the top gate **310A** is applied to the top gate **310B** to adjust the weight of the semiconductor device A. In the above example, 0V is applied to the top gate **310B**.

To prevent the FN tunneling in the semiconductor device B, the potential of the bottom gate **330B** of the semicon-

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ductor device B is controlled to the same potential as the top gate **310B** of the semiconductor device B. In the above example, the control circuit controls the voltage of the bottom gate **330B** to 0V through the bottom gate line **2335**.

Then, an electric field from the top gate **310B** to the bottom gate **330B** is not formed in the semiconductor device B, thereby inhibiting the weight adjustment.

The semiconductor device C is a synapse cell that shares the bottom gate line **1335** with the target semiconductor device A. The bottom gate **330C** of the semiconductor device C is already applied with the program voltage of the bottom gate **330A** of the semiconductor device A for adjusting the weight of the semiconductor device A. In the above example, 12V is already applied to the bottom gate **330C**.

The FN tunneling of the semiconductor device C is prevented by boosting the channel of the semiconductor device C under the influence of the program potential difference (or the program voltage (V_{PP})) of the semiconductor device A. In the above example, the control circuit **30** applies a program inhibit voltage to the top gate **310C** through the top gate line **2315** to prevent the movement of electric charge.

The program inhibit voltage applied to the top gate **310C** may be, for example, approximately half of the program voltage (V_{PP}), such as $0.4 V_{PP}$ to $0.6 V_{PP}$ but is not limited thereto.

Accordingly, the potential difference between the bottom gate **330C** and the top gate **310C** reduces, thereby preventing electric charge of the top gate **310C** from moving to the charge storage layer **430C**, and as a consequence, inhibiting the weight setting in the semiconductor device C.

The semiconductor device D is a synapse cell that shares the top gate line **2315** with the semiconductor device C and the bottom gate line **2335** with the semiconductor device B. Accordingly, the top gate **310D** of the semiconductor device D already has the same voltage as the program inhibit voltage of the top gate **310C**, and the bottom gate **330D** has the same voltage as the voltage of the bottom gate **330D** of the semiconductor device D.

In some embodiments, the semiconductor device D may have a potential difference between the gates **310D**, **330D** avoiding FN tunneling. In this case, the semiconductor device D is automatically inhibited from programming by the voltage control of the previous semiconductor devices A, B, C.

In some other embodiments, the semiconductor device D may have a potential difference between the gates **310D**, **330D** leading to FN tunneling. However, in this case, FN tunneling occurs from the bottom gate **330D** to the top gate **310D**, and as the top gate **310D** does not have the charge storage layer **430**, weight setting does not occur. Since in general, flash memory performs the erase operation before the program operation, electric charge stored in the charge storage layer **430** does not move to the top gate **310D** by the above-described potential difference.

Subsequently, the synapse weight may be independently adjusted for the remaining semiconductor devices B, C, D within the synapse block **10** via a process that includes the step of applying the program voltage (V_{PP}) to the bottom gate **330** of the new target semiconductor device B, C, or D and the step of applying the program inhibit voltage (for example, $0.5V_{PP}$) to the top gate **310** of the other semiconductor device B, C, or D sharing the bottom gate line **335** of the new target semiconductor B, C, or D.

The erase process of erasing the synapse weight already set to the synapse block **10** of FIG. **8** is similar to the above-described synapse weight setting process. However,

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the erase process may be performed by applying an electric field of the opposite direction to the electric field for setting the synapse weight to the target synapse cell as described above in the program/erase operation of the semiconductor device. For example, the synapse weight may be erased by forming an electric field of the opposite direction by a voltage having the same absolute value as the program voltage for weight setting but the opposite symbol.

FIG. 9 is a diagram showing the characteristics of the target semiconductor device of FIG. 8 in the synapse weight setting process.

In the graph of FIG. 9, x axis indicates the voltage of the bottom gate 310A, and y axis indicates the intensity of electric current of the gate terminal 310A, 330A. The intensity of electric current is the absolute value.

Experimentally, after allowing the source 210A and the drain 230A of the semiconductor device A in the synapse block 10 to float, 0V is applied to the top gate 310A, and the electric current of the top gate 310A and the bottom gate 330A is measured with varying voltage of the bottom gate 330A from 0V to 14V. As a result, it is found that gate currents having the same magnitude and the opposite symbols flow in the two terminals 310A, 330A.

That is, the graph of FIG. 9 shows that the tunneling current formed toward the bottom gate 330A is based on tunneling from the top gate 310A.

Additionally, the magnitude of the tunneling current of the semiconductor device A is on 10 nA level. The magnitude of the tunneling current is a very small value, compared to the weight setting current of the existing NOR flash memory. Accordingly, the graph of FIG. 9 supports that when the semiconductor device and the synapse block according to the embodiments of the present disclosure performs the program operation through FN tunneling, it is possible to reduce power consumption to at least one thousandth of the CHI process used in the existing NOR flash memory.

FIG. 10 is a diagram showing the result according to the synapse weight setting method of FIG. 8.

By the method for setting the synapse weight of the target semiconductor device described with reference to FIG. 8, the synapse weight of the target semiconductor device A may be determined. The synapse weight setting in the semiconductor device is represented as a shift of the threshold voltage of the semiconductor device.

As shown in FIG. 10, only the threshold voltage of the semiconductor device A intended to set the synapse weight is shifted to the right, and the threshold voltage of the remaining semiconductor devices B, C, D inhibited from programming is almost maintained without any substantial change.

While the present disclosure has been hereinabove described with reference to the embodiments shown in the drawings, this is provided for illustration purposes only and it will be appreciated by those skilled in the art that various modifications and variations may be made thereto. However, it should be noted that such modifications fall in the technical protection scope of the present disclosure. Therefore, the true technical protection scope of the present disclosure should be defined by the technical spirit of the appended claims.

INDUSTRIAL APPLICABILITY

With the Fourth Industrial Revolution, artificial intelligent related industry is globally growing very fast, and in par-

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ticular, many studies are being made on electronic devices that emulate the functions of synapses and neurons of the biological nervous system.

The semiconductor device according to the embodiments of the present disclosure can provide a synapse array (and a synapse block) with compatibility with the existing semiconductor process and high reliability. Accordingly, it is expected to provide high competitiveness in the field of neural network technology in hardware aspect.

The invention claimed is:

1. A semiconductor device, comprising:

a body made of a first conducting semiconductor material; a source and a drain that are made of a second conducting semiconductor material formed on the body; a gate insulating layer formed on the body; a first gate formed on the gate insulating layer; a second gate formed opposite the first gate, wherein the body is positioned between the first gate and the second gate; and

an insulating layer stack having a charge storage layer formed between the body and the second gate;

wherein electrons or holes of the first gate tunnel into the second gate on a state where the source and the drain are floated if an electric field from the first gate to the second gate is formed by a potential difference between the first gate and the second gate.

2. The semiconductor device according to claim 1, wherein the insulating layer stack further comprises:

a first stack insulating layer formed on the charge storage layer; and

a second stack insulating layer formed opposite the first stack insulating layer,

wherein the charge storage layer is positioned between the first stack insulating layer and the second stack insulating layer.

3. The semiconductor device according to claim 2, wherein the first stack insulating layer is configured to allow electric charges of the first gate to tunnel to the second gate if the electric field is formed by a potential difference between the first gate and the second gate, wherein the potential difference between the first gate and the second gate causes to maintain each insulating layer without damaging each insulating layer.

4. The semiconductor device according to claim 3, wherein the first stack insulating layer has a thickness of less than 7 nm.

5. The semiconductor device according to claim 3, wherein the first stack insulating layer is further configured to prevent the electric charge tunneled in the charge storage layer from leaking when the electric field disappears.

6. The semiconductor device according to claim 5, wherein the first stack insulating layer has a thickness of 3 nm or more.

7. The semiconductor device according to claim 2, wherein the second stack insulating layer is configured to inhibit tunneling of electric charges of the first gate to the second gate if an electric field is formed by a potential difference between the first gate and the second gate, wherein the potential difference between the first gate and the second gate causes to maintain each insulating layer without damaging each insulating layer.

8. The semiconductor device according to claim 7, wherein the second stack insulating layer has a thickness of 8 nm or more.

9. The semiconductor device according to claim 1, wherein the body is made of any one of a polycrystalline

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semiconductor material, an amorphous semiconductor material and a combination thereof.

10. A neural network having a plurality of semiconductor devices of claim **1** as synapse cells, the neural network comprising:

a synapse block including a first synapse array and a second synapse, wherein the first synapse array includes a first semiconductor device and a second semiconductor device and a second synapse array includes a third semiconductor device and a fourth semiconductor device, each synapse array configured such that drains of different semiconductor devices are arranged adjacent to each other;

drain lines electrically connected to a drain region of each synapse array;

first gate lines electrically connected to a first gate of each synapse array; and

source lines electrically connected to a source region of the first synapse array and a source region of the second synapse array,

wherein the drain line and the first gate line receive an input signal from a pre-neuron, and the source line outputs a calculation result at the synapse block as an output signal to a post-neuron.

11. The neural network according to claim **10**, wherein the synapse block is configured to: form a source signal based

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on the input signal and a preset weight for each semiconductor device when receiving the input signal.

12. The neural network according to claim **11**, wherein the output signal is a sum of the source signals of each semiconductor device.

13. The neural network according to claim **10**, wherein the synapse block connects the pre-neuron and the post-neuron through first switching devices which switch on/off electrical connections of the pre-neuron, the post-neuron and the synapse block, and

the synapse block is further connected to a control circuit through a second gate line electrically connected to a second gate of the first synapse array and a second gate of the second synapse array; and a second switching device which switches on/off an electrical connection between the control circuit and the synapse block, wherein the control circuit controls the semiconductor device through at least some of the plurality of lines.

14. The neural network according to claim **10**, wherein the output of the calculation result is performed in an ON state of the first switching device and an OFF state of the second switching device, and

the operation of the control circuit is performed in an OFF state of the first switching device and an ON state of the second switching device.

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